Biodegradable plastic (Poly–β–Hydroxybutyrate) production by marine microorganisms isolated from Tamilnadu coastal area

EXECUTIVE SUMMARY OF UGC-MRP FINAL REPORT

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Submitted by

Dr. A. Arun

Associate Professor and Head (i/c)

Department of Microbiology

Alagappa University

[Accredited with A+ Grade by NAAC (CGPA: 3.64) in Third Cycle,

Graded as Category-I University and Granted Autonomy by MHRD-UGC]

Karaikudi – 630 003, Tamilnadu, India.

Submitted To

University Grants Commission,

Ministry of Human Resource Management,

Govt. of India

Objective of the Project:

- To isolate and identify the PHB producing bacteria from the marine arenas of Tamilnadu such as Ennore, Tondiarpet, Triplicane, Adyar Estuary, Neelankarai, Muttukadu Backwaters, Mahabalipuram, Kalpakkam , Puducherry, Cuddalore, Parangipettai, Karaikal, Velankanni, Palk Strait, Thondi, Keelakarai, Tuticorin, Kulasekarapatnam and Kanyakumari.
- To compare the PHB productivity by the isolates and *Wausteria eutropha* (MTCC 1285)
- To overcome the problem associated with high production cost, in the present study, steps will be taken to evaluate the effect of carbon sources, nitrogen sources, pH and sodium chloride concentration on PHB productivity by the marine isolates
- To do fermentative production of PHB by the selected higher PHB producer using 3L Fermenter in the lab scale
- To quantitatively analyze the PHB produced during the study using UV-VIS Spectrophotometry; qualitatively analyze by Fourier Transmission Infra-red Spectrophotometry (FTIR analysis) and ^{1}H NMR Spectral analysis.

Summary of the findings:

- Samples were collected from from the marine arenas of Tamilnadu such as Ennore, Tondiarpet, Triplicane, Adyar Estuary, Neelankarai, Muttukadu Backwaters, Mahabalipuram, Kalpakkam, Puducherry, Cuddalore, Parangipettai, Karaikal, Velankanni, Palk Strait, Thondi, Keelakarai, Tuticorin, Kulasekarapatnam and Kanyakumari for the isolation of marine bacterial PHB producers.
- The PHB producing ability of the culture was checked by Sudan black B staining (Kitamura and Doi, 1994).
- The PHB producing marine microorganisms 13 were subjected to characterization according to the guidelines of Bergeys manual of Bacteriology (Holt, 1997) and 16 S rRNA sequencing and thus the marine isolates were identified.
- *Micrococcus luteus* with the potential for accumulating PHB was isolated from marine benthic environments. The various bioprocess parameters and media constituents were optimized for enhanced PHB production under different conditions
- The PHB produced were extracted (Ramsay et al., 1994) and estimated by using UV Spectrophotometer (Lee et al., 1995)
- The structural and dynamic properties of PHB were determined by UV and FTIR ^{1}H NMR (Castillo et al., 1986).
- Present study demonstrated the role of different carbon substrate, nitrogen and different pH levels of PHB production was studied.
- A significantly higher maximum biomass of 5.12g/L of PHB production was obtained using glucose as carbon, ammonium sulphate as nitrogen source at pH 7.

Batch experiments of the culture *Micrococcus luteus* was studied in a 3L fermenter containing 2L media**.** Initial pH of the media was adjusted to pH 7.0. Maximum biomass (21.5g/L) and PHB concentration (12.2g/L) were obtained at the 64th hour of fermentation.

Output/Achievement of the project

Manpower trained: **1 Person (Project Fellow)**

Ph.D Enrolled: **Mr. K. Mohan Rasu (Reg. No. 0796, Alagappa University, Karaikudi)**

Publication from the project:

1. Mohanrasu, K., Premnath, N., Siva Prakash, G., Sudhakar, M., Boobalan, T., **Arun, A.** (2018). Exploring multi potential uses of marine bacteria; an integrated approach for PHB production, PAHs and polyethylene biodegradation. Journal of Photochemistry and Photobiology B: Biology. 185: 55-65. (**Impact Factor: 3.165**).
2. M. Sudhakar, S. Ozgur, T. Boobalan and **A. Arun** (2018). Biopolymer Synthesis and Biodegradation. In Book: Sustainable Biotechnology- Enzymatic Resources of Renewable Energy. Om V. Singh • Anuj K. Chandel (Edt). Springer International Publishing AG, Cham, Switzerland. ISBN 978-3-319-95479-0.
3. K. Mohanrasu, Rathinam Raja and **A. Arun** / Ed: Se-Kwon Kin - Marine Microbes: Potential Candidates in Bioremediation Process. Book Title: Encyclopaedia of Marine Biotechnology. Willey – Blackwell, 2017. ISBN: 978-1-119-14377-2. (Accepted).

Contribution to the Society:

- Production of marine microbial PHB will be a mile stone in making of pollution free environment by replacing non-biodegradable plastics.
- Production of cost minimized biodegradable bioplastics (PHB) leads to the development of new materials with potential applications in agricultural, biomedical and environmental field.